Project Proposal: AI-Driven Stock Market Simulation and Trading App

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The capstone project I wanted to work on is creating an app that actively keeps track of the stock market, using that data to train ai models based on patterns within the market. The app will provide visual representation of market data, agent performance, and back testing capabilities. Many python libraries and separate APIs can keep track of the market.

The project goals are:

- Develop a Functional App: Create a Python application that simulates a simplified stock market environment.
- Implement AI Agents: Design and implement at least three distinct AI agents with varying trading strategies (e.g., Moving Average, RSI, Deep Learning-based Prediction).
- Data Integration: Integrate real-time or historical stock market data for training and testing agents.
- Performance Evaluation: Establish metrics and visualizations to evaluate agent performance and compare strategies.
- Back testing Capabilities: Implement a back testing feature to evaluate strategies on past market data.

Some of the deliverables are:

- Deliverable 1: Baseline AI Agent & Simulated Market: A functional application with a basic simulated stock market and at least one AI agent (Moving Average Strategy) capable of making trading decisions and tracking portfolio performance. Includes initial visualizations of agent behavior.
- Deliverable 2: Multiple AI Agents & Basic Back Testing: The application incorporates at least two additional AI agents (RSI & Simple RNN), along with a simplified back testing feature allowing for evaluation of strategies on historical data. Performance metrics will be established and visualized.
- Deliverable 3: Complete Application with Visualizations & Documentation: A fully functional application incorporating all AI agents, comprehensive visualizations (portfolio value, agent actions, market trends), and clear documentation of the code and project design. Includes a presentation summarizing findings.